

	Case Name: Asti-Cuneo Highway	Sector	Construction (civil)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Anna Bierma		One of nine std. scenarios
Date	2012		User defined distributions
File Name	C2012-04 Asti-Cuneo Highway	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

The construction of a new one mile long road section of the A33 highway Asti-Cuneo in Italy, including an overpass and a subway crossing.

The project consists of activity and cost data that were obtained directly from the actual project owner and resource data that were created by the user.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	67	Serial/Parallel (SP)	63%
Planned Duration (PD)	284d	Activity Distribution (AD)	54%
Budget At Completion (BAC)	6,230,961 €	Length of Arcs (LA)	0%
Renewable Resources	16	Topological Float (TF)	10%
Consumable Resources	2		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	8.5	11.5	3.4
CRI-rho	14.3	17.1	1.7
CRI-tau	21.6	30.2	2.1

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	21.7	37.4	1.4
SI	36.8	33.9	0.9
SSI	2.6	7.9	4.5
CRI-r	9.5	10.1	2.2
CRI-rho	9.5	10.1	2.2
CRI-tau	10.3	7.1	1.1

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	7.2	22.1	3.3
CRI-rho	7.2	22.1	3.3
CRI-tau	7.2	22.1	3.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	2.59	0.25
PV - SPI	2.98	0.57
PV - SCI	3.57	0.77
ED - 1	3.18	0.92
ED - SPI	2.96	0.55
ED - SCI	3.33	0.60
ES - 1	1.39	-0.47
ES - SPI(t)	1.75	-0.30
ES - SCI(t)	2.19	-0.24

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.4	-0.2
CPI	1.0	0
SPI	1.2	0.1
SPI(t)	1.2	0.0
SCI	2.0	0.2
SCI(t)	2.0	0.1
0.8 CPI + 0.2 SPI	1.0	0.0
0.8 CPI + 0.2 SPI(t)	1.0	0.0

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

The user has not performed any project control and therefore no tracking periods have been defined. Tracking periods can now be generated automatically by ProTrack or by manually inputting tracking data period by period..

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)